Tutorial 1: Introduction to Matlab

MAT3007 Optimization, 2023 Summer

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6.5-6.9

Outline

1. Matlab and CVX
2. Optimization Examples
3. Exercise

What is Matlab

- **Matlab** is an abbreviation for "matrix laboratory."
- **Matlab** is in automobile active safety systems, interplanetary spacecraft, health monitoring devices, smart power grids, and LTE cellular networks.
- The matrix-based **Matlab** language is the world's most natural way to express computational mathematics.

Comparison for defining a matrix

Consider the matrix

$$A = \begin{bmatrix} 9 & 5 \\ 7 & 4 \end{bmatrix}$$

Language	Define a Matrix	$A^T A$
Python	A = [[9,5], [7,4]]	Hard to implement
Python with Numpy	A = numpy.array([[9,5], [7,4]])	np.multiply(A.transpose(), A)
R with Numpy	A < -cbind(c(9,7), c(5,4))	t(A)*A
Matlab	A = [9,5; 7,4]	A' * A

Table: Illustration on different programming languages' matrix operation.

Matlab Installation

1. Create a "Mathworks" account

- Go to the webpage `https://mathworks.com/`.
- Create a account. **Use the university's email**, which is ended with `link.cuhk.edu.cn`.
- Verify the account and fill up personal information. Once finished, you will see a page as below, indicating the individual lincense has been created.

License	Label	Option
40864163	MATLAB (Individual)	Total Headcount

Figure: Individual License

Matlab Installation

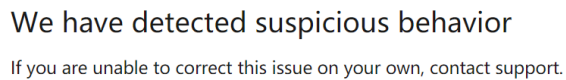
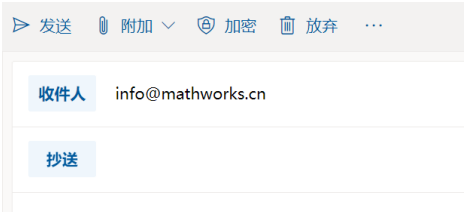


Figure: Account Creation Issue

- If you have the account creation issue as shown in the picture, send a email to **info@mathworks.cn** with your university's email.



Matlab Installation

2. Install the Matlab

- Download the Matlab
  - **In campus**: `https://cuhko365.sharepoint.com/sites/swcenter`
  - **Out-of campus**: `https://www.mathworks.com/downloads`
- Install(You may need to **close the firewall**.)
  - **Windows**: Unzip and click setup.exe.
  - **MacOS**: Double click InstallForMacOSX.app
  - **Linux**: Execute the installation program: `./install`
- Activate

Matlab Basics

- Basic operations: $+, -, /, *$.
- Matrix operations:
  - `A=[1,2;3,4]` $\rightarrow A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
  - Identity matrix: eye(3) $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
  - Zeros matrix: zeros(size)
  - Random matrix: randn(size)
  - Matrix multiplication: $A * B$
  - Element-wise product: $A. * B$
  - Transpose: A'
- Functions
  - sum
  - max
  - abs
  - inv

What is CVX

- CVX is a Matlab-based modeling system for convex optimization.
- CVX allows constraints and objectives to be specified using standard Matlab expression syntax.

CVX Installation

1. Retrieve the latest version of CVX standard bundles from `http://cvxr.com/cvx/download/` site; (.zip is recommended)
2. Unpack the file; a directory called cvx will be created.
3. Start Matlab.
4. Change directories to the top of the CVX distribution, and run "cvx_setup" command in terminal.

CVX Example

Consider the linear programming problem

$$\begin{aligned} \max \quad & 2x + 3y \\ & x + y \leq 8 \\ & 2x + y \leq 15 \\ & x \leq 6 \\ & y \leq 4 \\ & x, y \geq 0 \end{aligned}$$

We can use the following Matlab CVX program to solve it

```
cvx_begin
    variable x(1)
    variable y(1)
    maximize (2*x + 3*y)
    subject to
        x + y <= 8
```

Useful sources

Matlab

- CUHKSZ Matlab installation guide: `https://mp.weixin.qq.com/s/dgDN4f6JKN3Zjl3GTrJh6g`

CVX

- CVX installation guide: `http://cvxr.com/cvx/doc/install.html`
- More CVX examples: `http://cvxr.com/cvx/examples/`

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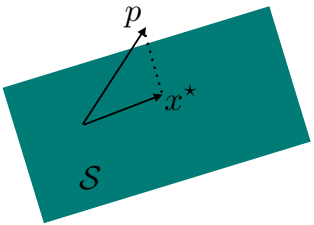
Obtain Problems

- **Decision Varibles**: the problem is to decide what value each variable should take
- **Objective Function**: the objective function indicates how much each variable contributes to the value to be optimized in the problem
- **Constraints**: constraints define the possible values the variables of a linear programming problem may take

Optimization in Geometry

Projection

- Project a point $p \in \mathbb{R}^n$ to a plane (or any set) S in $\mathbb{R}^d$.
- Equivalent to find the minimum distance from p to any point $x \in S$.



$$\begin{aligned} \min_{x} \quad & ||p - x||^2 \\ s.t. \quad & x \in S \end{aligned} \tag{1}$$

Optimization in Combinatorial Problem

Knapsack problem

- you are a thief, you break in a room and see N items.
- the weights and values of each items are w_i and v_i.
- the capacity of your knapsack is W.
- decide which items to take
- let $x_i \in \{0, 1\}$ represent whether or not to take the i-th items.

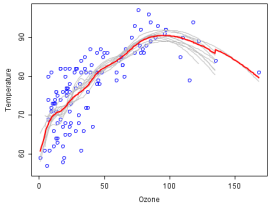


$$\begin{aligned} \max_{x} \quad & \sum_{i=1}^{N} x_i v_i \\ s.t. \quad & \sum_{i=1}^{N} x_i w_i \leq W \\ & x_i \in \{0, 1\}, \quad i = 1, \cdots, N \end{aligned} \tag{2}$$

Optimization in Machine learning

(General) Regression

- you have a dataset $\{x_i, y_i\}_{i=1}^N$.
- $x_i \in \mathbb{R}^d$ are your features.
- $y_i \in \mathbb{R}$ are your labels.
- train a regression model $f_\theta(x)$ to "fit" the data.
- let $f_\theta(x_i) \approx y_i$, $\forall i \in [N]$.



$$\min_{\theta} \quad \frac{1}{N}\sum_{i=1}^{N}\left(f_\theta(x_i) - y_i\right)^2 \tag{3}$$

Optimization in Machine learning

Linear Regression

- you choose the linear model $f_\theta(x_i) = \theta^T x_i + \theta_0$, $\theta \in \mathbb{R}^d$.

$$\min_{\theta,\theta_0} \quad \frac{1}{N}\sum_{i=1}^{N}(\theta^T x_i + \theta_0 - y_i)^2. \tag{4}$$

- in matrix form

$$\min_{\theta} \quad \frac{1}{N}||X\theta - Y||^2$$

where $\theta \in \mathbb{R}^{d+1}, X \in \mathbb{R}^{N\times(d+1)}, Y \in \mathbb{R}^N$.

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Exercise: Optimization in Portfolio Management

Markowitz Mean-variance model (1952)

You want to invest your money to N securities:

- the returns of the securities are $\boldsymbol{r} = [r_1, r_2, \cdots, r_N]^T$.
- the expected return of the securities are $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{r}]$.
- the covariance between 2 securities are $Cov(r_i, r_j)\ \forall i, j \in [N]$.
- the covariance matrix is defined by

$$\Sigma = \begin{bmatrix} Var(r_1) & \cdots & Cov(r_1, r_N) \\ \vdots & & \vdots \\ Cov(r_N, r_1) & \cdots & Var(r_N) \end{bmatrix}$$



- the *proportion* you invest to each securities is $\boldsymbol{x} = [x_1, \cdots, x_N]^T$.

Exercise: Optimization in Portfolio Management

- 1. What is the mean and variance of your portfolio?
- 2. If you want to maximize your return, and control your risk by T_{max}, how to formulate the optimization problem?
- 3. If you want to minimize your risk, and maintain a least expected return R_{min}, how to formulate the optimization problem?
- 4. If you want to **balance** your return and risk, how to formulate the optimization problem?(Hint: Introduce a hyperparameter λ to represent the trade off between return and risk).

Solution to Exercise

- 1. What is the mean and variance of your portfolio?
  Solution: $(\boldsymbol{x}^T\boldsymbol{\mu}, \boldsymbol{x}^T\Sigma\boldsymbol{x})$. □

- 2. If you want to maximize your return, and control your risk by T_{max}, how to formulate the optimization problem?
  Solution:
  1. Decision variable: proportion x.
  2. Objective function: return $x^T\mu$.
  3. Constraints: control the risk; property of proportion.

$$\begin{aligned} \max_{\boldsymbol{x}} \quad & \boldsymbol{x}^T\boldsymbol{\mu} \\ s.t. \quad & \boldsymbol{x}^T\Sigma\boldsymbol{x} \leq T_{max} \\ & \boldsymbol{x}^T\mathbf{1} = 1, \quad \boldsymbol{x} \geq 0 \end{aligned} \tag{5}$$

□

Solution to Exercise

3. If you want to minimize your risk, and maintain a least expected return R_{min}, how to formulate the optimization problem?

Solution:

1. Decision variable: proportion x.
2. Objective function: risk $x^T \Sigma x$.
3. Constraints: maintain least return; property of proportion.

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{x}^T \Sigma \boldsymbol{x} \\ s.t. \quad & \boldsymbol{x}^T \boldsymbol{\mu} \geq R_{min} \\ & \boldsymbol{x}^T \mathbf{1} = 1, \quad \boldsymbol{x} \geq 0 \end{aligned} \tag{6}$$

□

Solution to Exercise

4. If you want to **balance** your return and risk, how to formulate the optimization problem?(Hint: Introduce a hyperparameter λ to represent the trade off between return and risk).

Solution:

1. Decision variable: proportion x.
2. Objective function: combination of return and risk loss, parameterized by λ.
3. Constraints: property of proportion.

$$\begin{aligned} \max_{\boldsymbol{x}} \quad & \boldsymbol{x}^T\boldsymbol{\mu} - \lambda \boldsymbol{x}^T\Sigma\boldsymbol{x} \\ s.t. \quad & \boldsymbol{x}^T\mathbf{1} = 1, \quad \boldsymbol{x} \geq 0 \end{aligned} \tag{7}$$

where λ represents the trade off between return and risk.

1. if λ is small, the primary goal is to maximize the return while the effect of risk is negligible.
2. if λ is large, the contribution of return to objective function is small, so it's essentially minimizing risks.

□

Acknowledgements

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